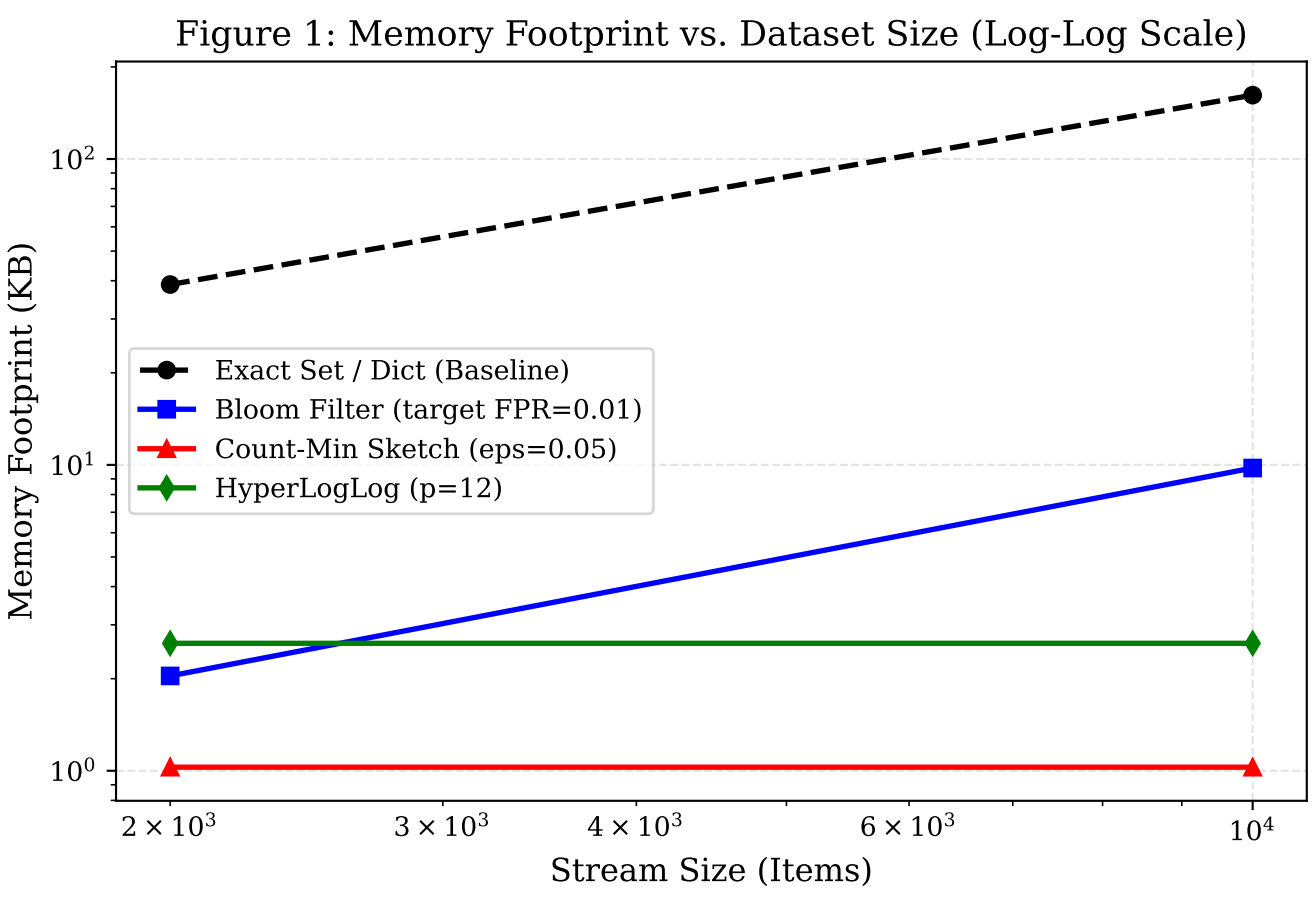
Figure 1: Memory Footprint vs. Dataset Size (Log-Log Scale)
Memory Footprint (KB)
10^2
10^1
10^0
Exact Set / Dict (Baseline)
Bloom Filter (target FPR=0.01)
Count-Min Sketch (eps=0.05)
HyperLogLog (p=12)
2×10^3
3×10^3
4×10^3
6×10^3
10^4
Stream Size (Items)